

MA378 Chapter 3: Numerical Integration**§3.1 Introduction to Newton-Cotes
Quadrature: The Trapezium Rule**

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Tablet was not working, so all examples,
derivations and calculations were done on the board.
And then some notes were added (long)
after the class.

1.0 We'll run the rule over the following topics:

1 Introduction

2 Newton-Cotes methods

3 The Trapezium rule

- Method 1
- Method 2
- Method 3

4 Exercises

1.1 Introduction

Problem

Given a real-valued function f that is continuous on $[a, b]$, can we find an estimate for

$$I(f) := \int_a^b f(x)dx?$$

And if we can, can we say how accurate it is?

1.1 Introduction

Why is this an interesting problem?

- ▶ Many problems in applicable mathematics require **definite integrals** to be evaluated. These methods were originally motivated by problems in such varied areas as astronomy, probability, and brewing. They are now ubiquitous.
- ▶ Evaluating them by finding the anti-derivative can be hard, and very hard to automate.
- ▶ Sometimes, although the function is integrable, its anti-derivative doesn't exist in a closed form.

1.1 Introduction

The process of numerically estimating a definite integral is called **Numerical Integration** or **Quadrature**.

The formulae we'll derive all look like

$$Q_N(f) := q_0 f(x_0) + q_1 f(x_1) + q_2 f(x_2) + \cdots + q_N f(x_N).$$

Here the points x_i are called **quadrature points** and the q_i are **quadrature weights**.

We need a way of choosing these.

The simplest approach for choosing the quadrature points is to take them to be equally spaced, i.e., $x_i = a + hi$ where $h = (b - a)/N$.

1.1 Introduction

How to choose the weights?

We've spent a lot of time on approximating functions with polynomials. So it is natural to compute a polynomial interpolant to f , and take its integral. The appeal of this approach is due to the fact that

- ▶ We know how to compute polynomial interpolants.
- ▶ Integrating polynomials is easy.
- ▶ We can estimate the error easily (yet again, we'll make use of Cauchy's Theorem).

This leads to the **Newton-Cotes** methods, which are the subject of this section, and the next one. Later again, we'll look at more sophisticated methods, called **Gaussian Methods** which use non-uniformly spaced points.

1.2 Newton-Cotes methods

Definition 1.1 (Newton-Cotes quadrature)

The **Newton-Cotes** quadrature rule for $\int_a^b f(x)dx$ with $N + 1$ points is derived by integrating exactly the polynomial of degree N that interpolates f at the $N + 1$ equally spaced points $a = x_0 < x_1 < \cdots < x_N = b$. The method is written as

$$Q_N(f) := q_0 f_0 + q_1 f_1 + q_2 f_2 + \cdots + q_N f_N,$$

where we use the notation $f_k := f(x_k)$.

That is, the quadrature weights are chosen so that

$$Q_N(f) = \int_a^b p_N(x)dx,$$

where p_N is the polynomial of degree N that interpolates f at the $N + 1$ quadrature points...

1.2 Newton-Cotes methods

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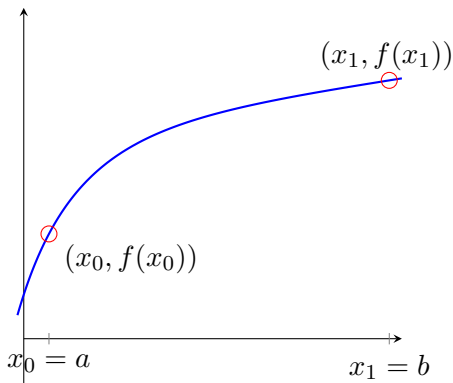
where p_N is the polynomial of degree n that interpolates f at the $N + 1$ quadrature points.

However, it turns out that we can compute the weights $q_0, q_1, \dots, q_N$, **without** knowing p_N .

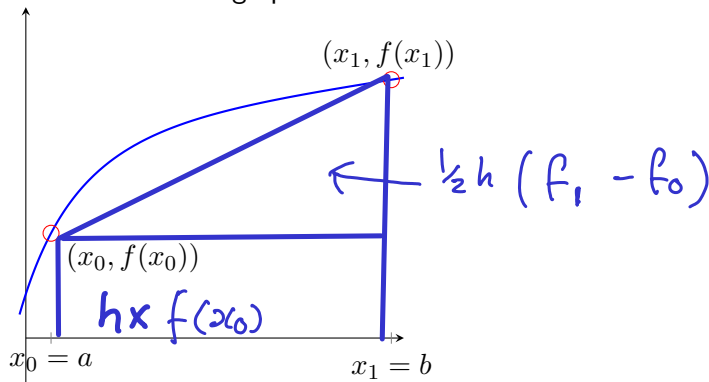
We'll do this for $N = 1$ in the next section, and $N = 2$ (the most interesting case) in Section 3.2.

1.3 The Trapezium rule

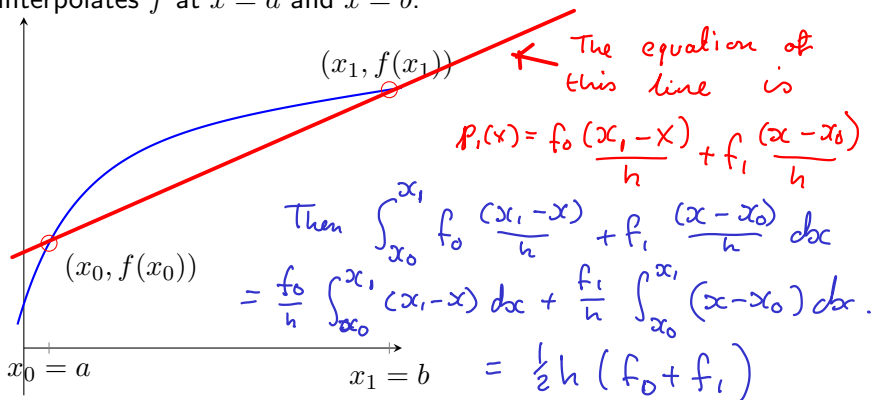
Suppose we wanted to estimate the integral of a function, f , shown below, on the interval $[a, b]$.



Method 1: We could try to estimate the area of the trapezium that fits under the graph:



Method 2: We could find p_1 , the polynomial of degree 1 that interpolates f at $x = a$ and $x = b$:



Note that this shows that $q_i = \int_a^b L_i(x) dx$, where, as usual, the L_i are the Lagrange Polynomials.

Method 3: The third approach for generating the Trapezium Rule is called the **Method of Undetermined Coefficients**. Because the method is based on integrating a linear function we expect it to yield an exact solution for any constant or linear function (i.e., there should be no error). To keep the algebra simple, we'll take $a = 0$ and $b = 1$. So,

$$Q_1(f) = q_0 f(0) + q_1 f(1),$$

and, setting $f(x) \equiv 1$, and then $f(x) = x$ we get

$$f \equiv 1 \quad \int_0^1 f(x) dx = 1 \quad \& \quad Q_1(f) = q_0 + q_1$$

so $q_0 + q_1 = 1$

$$f \equiv x \quad \int_0^1 x dx = \frac{1}{2}, \quad Q(f) = q_0(0) + q_1(1) \Rightarrow \begin{cases} q_1 = \frac{1}{2} \\ \Rightarrow q_0 = \frac{1}{2} \end{cases}$$

Now we need to extend this to estimating $\int_a^b g(x)dx$ as follows:

Define a mapping from $[0,1]$ to $[a,b]$ as

$$t = a + (b-a)x. \quad \text{Note: } \frac{dt}{dx} = b-a.$$

$$\text{Let } f(x) = g(\underbrace{a + (b-a)x}_t)$$

$$\text{So } \int_a^b g(x) dt = \int_0^1 f(x) (b-a) dx = (b-a) \int_0^1 f(x) dx.$$

$$\text{So } \underbrace{\frac{1}{2} f(0) + \frac{1}{2} f(1)} \rightarrow \frac{(b-a)}{2} (g(a) + g(b))$$

Trap Rule on $[0,1]$

Trap Rule on $[a,b]$

Example 1.2

Use the Trapezium Rule to estimate

$$\int_0^{\pi/4} \cos(x) dx. \quad \left\{ \begin{array}{l} \text{Error is } 0.0367 \end{array} \right.$$

Calculate the (exact) error $|\int_a^b f(x) dx - Q_1(f)|$.

$$f(x) = \cos(x), \text{ and } a = x_0 = 0, \quad b = x_1 = \pi/4.$$

$$\text{Also } h = \pi/4$$

$$\begin{aligned} \text{Then } Q_1(f) &= \frac{b-a}{2} (f(a) + f(b)) = \frac{\pi/4}{2} (\cos(0) + \cos(\pi/4)) \\ &= 0.67038. \end{aligned}$$

$$\text{Also } I(f) = \int_0^{\pi/4} \cos(x) dx = \sin(x) \Big|_0^{\pi/4} = \frac{1}{\sqrt{2}} = 0.7071$$

1.4 Exercises

Exercise 1.1

(For simplicity, you may assume that the quadrature rule is integrating f on the interval $[-1, 1]$.) Let $q_0, q_1, \dots, q_N$ be the quadrature weights for the Newton-Cotes rule $Q_N(f)$. Show that $q_i = q_{N-i}$ for $i = 0, \dots, N$.

Exercise 1.2

Show that $\sum_{i=0}^n q_i = b - a$.